

Experiment 'p_3_9_temperature' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_9_temperature

Natural language statement: Write a method that takes a temperature value and a letter C for Celsius or F for Fahrenheit. Return whether water is liquid, solid, or gaseous at the given temperature at sea level.

Method signature: p_3_9_temperature (t: real, c: char) returns (s: string)

Ensures

- `if c == 'C' then if t < 0.0 then s == "solid" else if t == 0.0 then s == "solid or liquid" else if 0.0 < t < 100.0 then s == "liquid" else if t == 100.0 then s == "liquid or gaseous" else s == "gaseous" else if t < 32.0 then s == "solid" else if t == 32.0 then s == "solid or liquid" else if 32.0 < t < 212.0 then s == "liquid" else if t == 212.0 then s == "liquid or gaseous" else s == "gaseous"`

Requires

- `(c == 'C' && t >= -273.15) || (c == 'F' && t >= -459.67)`
- `c == 'C' || c == 'F'`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes a temperature value and a letter C for Celsius or F for
→ Fahrenheit. Return whether water is liquid, solid, or gaseous at the given
→ temperature at sea level.

The signature should be:

`p_3_9_temperature (t: real, c: char) returns (s: string)`

The method should respect the following contract:

```
requires (c == 'C' && t >= -273.15) || (c == 'F' && t >= -459.67), requires c == 'C' || c
→ == 'F', ensures if c == 'C' then if t < 0.0 then s == "solid" else if t == 0.0
→ then s == "solid or liquid" else if 0.0 < t < 100.0 then s == "liquid" else if t
→ == 100.0 then s == "liquid or gaseous" else s == "gaseous" else if t < 32.0 then s
→ == "solid" else if t == 32.0 then s == "solid or liquid" else if 32.0 < t < 212.0
→ then s == "liquid" else if t == 212.0 then s == "liquid or gaseous" else s == "
→ gaseous"
```

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dmdqvm04i4sVl3YV7odeZcL1pAYlw

Model: gpt-5.5-2026-04-23

Created at: 1780485309

Finish reason: stop

Usage: 696 tokens in, and 597 tokens out

```
{
  if c == 'C' {
    if t < 0.0 {
      s := "solid";
    } else if t == 0.0 {
      s := "solid or liquid";
    } else if t < 100.0 {
      s := "liquid";
    } else if t == 100.0 {
      s := "liquid or gaseous";
    } else {
      s := "gaseous";
    }
  }
}
```

```

} else {
  if t < 32.0 {
    s := "solid";
  } else if t == 32.0 {
    s := "solid or liquid";
  } else if t < 212.0 {
    s := "liquid";
  } else if t == 212.0 {
    s := "liquid or gaseous";
  } else {
    s := "gaseous";
  }
}
}
}

```

Final Program

```

method p_3_9_temperature (t: real, c: char) returns (s: string)
  requires (c == 'C' && t >= -273.15) || (c == 'F' && t >= -459.67)
  requires c == 'C' || c == 'F'
  ensures if c == 'C' then if t < 0.0 then s == "solid" else if t == 0.0 then s == "
    ↪ solid or liquid" else if 0.0 < t < 100.0 then s == "liquid" else if t ==
    ↪ 100.0 then s == "liquid or gaseous" else s == "gaseous" else if t < 32.0
    ↪ then s == "solid" else if t == 32.0 then s == "solid or liquid" else if
    ↪ 32.0 < t < 212.0 then s == "liquid" else if t == 212.0 then s == "liquid or
    ↪ gaseous" else s == "gaseous"
{
  if c == 'C' {
    if t < 0.0 {
      s := "solid";
    } else if t == 0.0 {
      s := "solid or liquid";
    } else if t < 100.0 {
      s := "liquid";
    } else if t == 100.0 {
      s := "liquid or gaseous";
    } else {
      s := "gaseous";
    }
  } else {
    if t < 32.0 {
      s := "solid";
    } else if t == 32.0 {
      s := "solid or liquid";
    } else if t < 212.0 {
      s := "liquid";
    } else if t == 212.0 {
      s := "liquid or gaseous";
    } else {
      s := "gaseous";
    }
  }
}

```

```
}  
}  
}
```

Total Token Usage

Input tokens: 696

Output tokens: 597

Reasoning tokens: 354

Sum of ‘total tokens’: 1293

Experiment Timings

Overall Experiment started at 1780485309334, ended at 1780485324115, lasting 14781ms (14.78 seconds)

Iteration #1 started at 1780485309337, ended at 1780485324115, lasting 14778ms (14.78 seconds)